Semi-supervised learning

Tasks (Lab 13):

1. Consider any dataset corresponding to binary classsification. For example, you can generate synthetic dataset using function make_moons from scikit-learn library.

2. Split data into training and testing subsets.

3. Choose randomly g observations in the training data as labeled. Remove the labels for the remaining observations in the training data.

4. Selected Semi supervised learning methods are implemented in scikit-learn library: `https://scikit-learn.org/stable/modules/semi_supervised.html`

5. Compare 4 methods:

   (a) **Naive Method:** learn classifier using only g labelled examples in training data.

   (b) **Self-training**

   (c) **Label propagation**

   (d) **Label spreading**

   SSL methods (**Self-training**, **Label propagation** and **Label spreading**) use both labelled and unlabelled examples for training.

6. Use SVC (Support Vector Classifier) as a base classifier.

7. Compute accuracy, balanced accuracy, F measure and ROC AUC for the methods on testing data.

8. Analyse how the value of g affects the results.